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BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE 341: Microprocessors Semester: Summer 2026

Examination: Quiz 3 Duration: 45 Minutes Full Marks: 20

Name: _____ ID: _____ Section: _____

Question 1 [9 marks]

An 8086 system uses cascaded 8259A PICs for hardware interrupts. The system must handle **25 interrupt sources**. Due to faults, **IR3 and IR7** are unusable on all PICs. Slaves connect to the master sequentially from Master IR0, skipping faulty pins.

The values of the registers for executing the below code snippet:

SS = A000h, SP = FFFAh, CS = 3000h, DS = 2000h, ES = 1000h

The following instructions are executed sequentially:

Instruction 1:	MOV AX, EF17H
Instruction 2:	MOV BL, 13H
Instruction 3:	DIV BL
Instruction 4:	MOV VAR, AL

Assume that the interrupt service routine for the interrupt generated during execution of the above instructions starts at physical address **8B62AH**, with a fixed CS of **8A00H**.

a) Analyze with **proper justification** the given instructions and determine **which interrupt is generated** during their execution and its **interrupt type. [1 marks]**

b) Using the information provided about the ISR, calculate the **IP** of the ISR. Then determine the **four IVT memory addresses** and corresponding **hexadecimal data values** required to install this interrupt. **[2.5 marks]**

IP =	

IVT memory address	Data (in hex)

c) The instruction that generates the interrupt **starts at** physical address **31122H** and is **2 bytes long.** Deduce the hexadecimal values pushed onto the stack at physical addresses **AFFF4H through AFFF7H** when the interrupt occurs. Assume the **FLAGS** are pushed to **AFFF8H–AFFF9H. [3.5 marks]**

Address	**Data**

d) Determine the **minimum** number of PICs required to handle the **25 sources** including **master and slave(s). [2 marks]**

Question 2 [11 marks]

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a) An 8086-based system executes the following instruction:

MOV [DAD**X**H], **P**

Here, **P** is a source register and **X** is an unknown hexadecimal digit.
The memory subsystem is organized into even and odd memory banks, and the **A0** and **BHE'** pins are **both** reported to be **0(zero)** during the memory access.

(i) Determine a possible value of **X** that is consistent with the reported memory-bank selection. Justify your choice. **[1 marks]**

(ii) Recommend a suitable general-purpose register as the source of the MOV instruction. State whether the register should be 8-bit or 16-bit, and justify your answer based on the reported memory-bank selection. **[1 marks]**

b) The processor requires **two machine cycles** to complete the followign instruction:

MOV **P**, [FFF**Y**H]

Here, **P** is the register deduced in **question-(a)** and **Y** is an unknown hexadecimal digit. The CPU operates at **9 MHz,** and the peripheral memory requires **300 ns** to upload data on the data bus.

(a) Determine a possible value of **Y** based on the fact that the instruction requires **two machine cycles** to complete. Justify your choice and explain **why** two machine cycles are required for this particular operation. **[2 marks]**

(b) Explain with proper reasoning whether the processor will insert a **wait state** or not. **[1 marks]**

(c) Determine the **total time** required to execute the instruction: **MOV P, [FFFYH]**. **[2 marks]**

(d) Illustrate the timing diagram for the instruction. Include *CLK, AD0-AD15, A16-A19, ALE, RD'/WR', M/IO', READY* and *DEN'* pins in your diagram. **[4 marks]**

Question : Bonus [2.5 marks]

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3. Consider two **independent** events in the 8086 system that ultimately cause the processor to begin execution from the **same interrupt-service location associated with Type 2.** The two events originate through fundamentally different paths: one is initiated through the processor's **instruction-execution mechanism,** whereas the other is initiated through a dedicated **hardware interrupt-request mechanism** using the **INTR** pin.

a) Identify **sources/types** of the two interrupting events. **[0.5 marks]**

b) Distinguish their **triggering mechanism,** and **maskable/non-maskable nature**.**[1 marks]**

c) Explain how both the events reach the **same Type-2 ISR. [1 marks]**